Housing Reallocation and Demographic Transitions
A Two-Generation OLG Model

1 Analytical Model

This section isolates two implications of housing access in a two-period overlapping-generations economy. A financing-constrained young buyer can value family-sized space more than an old incumbent. A permanent change in desired fertility can also move the economy between positive steady states, generating house-price gains only during the transition. Appendix A gives the derivations and numerical construction.

1.1 Environment

Households. At date t, the economy contains Y_t young households and O_t old households. A young household has income y_i, liquid wealth b_i, and chooses total nondurable expenditure c, total housing h, completed fertility n, saving a', and tenure $m \in \{R, O\}$. The pair (y_i, b_i) is drawn from a fixed distribution F in each cohort; estates are warm-glow expenditures rather than the source of the next cohort's b_i.

Preferences. Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of housing. Define the parts of the household bundle available to the adults as

$$x = c - \chi n > 0, \qquad s = h - \kappa n > 0. \tag{1}$$

Young and old flow utilities are

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n, \tag{2}$$
$$u^2(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e. \tag{3}$$

Here e is a warm-glow estate. Both child requirements reduce the bundle left for adults. Logarithmic utility keeps conditional demands and the response to a binding housing limit in closed form; access constraints and tax timing carry the mechanisms.

Demography. A child born at t produces $\nu > 0$ young households at $t + 1$, after survival and household formation. Thus $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$, where $\bar{n}_t$ is average fertility among the young.

Housing and asset markets. The fixed housing stock is $\bar{H}$. Rental units satisfy $h \le h_R^{\max}$, owner units satisfy $h \le h_O^{\max}$, and $h_R^{\max} < h_O^{\max}$. The consumption good is the numeraire and trades through a one-period bond with price $q \in (0, 1)$ and gross return $R_f = q^{-1}$; only housing clears domestically. The owner of record pays the property tax. A competitive rental intermediary buys a unit for P_t at the beginning of date t, then collects r_t, pays $\tau^p P_t$, and holds a unit worth P_{t+1} at the end of the period. Free entry therefore imposes

$$r_t + P_{t+1} = R_f P_t + \tau^p P_t \quad \Longleftrightarrow \quad r_t = (R_f + \tau^p) P_t - P_{t+1}. \tag{4}$$

Rent is paid in date-$(t+1)$ goods, so its date-t value is qr_t; at a steady state, $r = (R_f - 1 + \tau^p)P$. Rental intermediaries are not taxed on resale in the baseline. Appendix A.5 shows that taxing their resale changes incidence but not the marginal-value comparison below.

Capital-gains taxation and home finance. Household capital gains are taxed only upon sale. An old owner who bought one unit at P_{t-1} and sells it at P_t pays the tax per unit

$$\ell_t = \tau^g \max\{P_t - P_{t-1}, 0\}, \tag{5}$$

where τ^g is the capital-gains tax rate. Retaining the house until death avoids this tax because the estate's basis is reset to the date-$(t+1)$ price; an immediate estate sale then realizes no taxable gain. Retained owner-occupied housing cannot be rented to another household. At death the estate sells it back into the fixed stock, and the proceeds enter warm-glow estate expenditure. A young buyer finances the share $\phi \in (0,1)$ and posts $(1-\phi)P_t h$ at purchase.

1.2 Household problems

Households take prices and transfers as given. Renters carry financial wealth into old age; owners carry the house and its mortgage.

Young renters. Conditional on renting, the young household solves

$$W_t^R(i) = \max_{c,h,n,a'} u_t^Y(c,h,n) + \beta V_{t+1}^R(R_f a') \tag{6}$$

$$\begin{aligned} \text{s.t.} \quad & c + a' + q\,r_t h = y_i + b_i + T_t, \\ & h \le h_R^{\max}, \qquad a' \ge 0. \end{aligned} \tag{7}$$

The renter enters old age with financial wealth $R_f a'$.

Young owners. Conditional on owning, the young household solves

$$\begin{aligned} W_t^O(i) = \max_{c,h,n,a'} \; & u_t^Y(c,h,n) + \beta V_{t+1}^O(a_{t+1}, h) \\ \text{s.t.} \quad & c + a' + [(1-\phi) + q\tau^p]P_t h = y_i + b_i + T_t, \\ & a_{t+1} = R_f a' - R_f \phi P_t h, \qquad (1-\phi)P_t h \le b_i, \qquad h \le h_O^{\max}, \qquad a' \ge 0. \end{aligned} \tag{8}$$

Closing occurs before current income and rebates are available, and unsecured borrowing is unavailable. Thus only liquid wealth b_i can satisfy the down payment. The transfer T_t is the date-t value of the equal rebate of housing-tax revenue.

Old households. An old household has financial wealth a. An owner also carries housing H from young age. Renters and owners solve, respectively,

$$V_t^R(a) = \max_{c^2,h^2,e} u^2(c^2,h^2,e), \qquad c^2 + qe + q\,r_t h^2 = a + T_t, \quad 0 < h^2 \le h_R^{\max}, \tag{9}$$

$$\begin{aligned} V_t^O(a,H) = \max_{c^2,h^2,e} u^2(c^2,h^2,e), \qquad & c^2 + qe + (q\,r_t - \ell_t)h^2 = a + (P_t - \ell_t)H + T_t, \\ & 0 < h^2 \le H, \quad e \ge P_{t+1}h^2. \end{aligned} \tag{10}$$

The marginal cost of retaining a unit is $q\,r_t - \ell_t$: selling realizes the tax, while retaining avoids it.

Tenure choice. Each household draws once an idiosyncratic preference ξ_i for ownership, independently of (y_i, b_i), and observes it before choosing tenure. It owns when $W_t^O(i) + \xi_i \ge W_t^R(i)$.

Assume ξ_i is logistic with location $\bar{\xi}$ and scale $\sigma_\xi > 0$, which gives the owner share

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}{\exp\{W_t^R(i)/\sigma_\xi\} + \exp\{[W_t^O(i) + \bar{\xi}]/\sigma_\xi\}}. \tag{11}$$

The taste draw smooths aggregate tenure shares without changing conditional renter or owner policies.

1.3 Housing constraints and the cost of children

An access constraint binds when desired housing exceeds the amount a household can rent or finance. Its shadow value is the willingness to pay for one more unit.

Fix tenure $m \in \{R, O\}$ and suppose saving and old-age choices are interior. Let $A = 1 + \gamma + \omega_B$ denote the sum of the old-age expenditure weights, and let lifetime resources, valued at date t, be

$$w_{it} = y_i + b_i + T_t + qT_{t+1}. \tag{12}$$

The current cost of housing depends on tenure:

$$u_t^R = q\, r_t, \qquad u_t^O = q(r_t + \ell_{t+1}). \tag{13}$$

A renter pays the present value of rent. A buyer who sells in old age also anticipates the gains tax. Eliminating saving and old-age choices gives

$$\begin{aligned} \max_{x,s,n} \quad & (1+\beta A)\log x + \alpha \log s + \vartheta_t \log n \\ \text{s.t.} \quad & (1+\beta A)x + u_t^m s + (\chi + \kappa u_t^m)n = w_{it}. \end{aligned} \tag{14}$$

The last budget term is the full resource cost of another child: χ units of goods plus κ units of housing valued at the tenure-specific user cost.

Without a housing-access constraint, define $D_t = 1 + \beta A + \alpha + \vartheta_t$. The household then chooses usable goods, usable housing, and fertility according to

$$x^m = \frac{w_{it}}{D_t}, \qquad s^m = \frac{\alpha w_{it}}{D_t u_t^m}, \qquad n^m = \frac{\vartheta_t w_{it}}{D_t(\chi + \kappa u_t^m)}. \tag{15}$$

Gross choices are $c^m = x^m + \chi n^m$ and $h^m = s^m + \kappa n^m$.

A renter cannot exceed the largest rental unit. A buyer cannot exceed either the largest owner unit or the amount supported by liquid wealth:

$$\widehat{h}_{it}^R = h_R^{\max}, \qquad \widehat{h}_{it}^O = \min\left\{h_O^{\max}, \frac{b_i}{(1-\phi)P_t}\right\}. \tag{16}$$

Below this limit, (15) applies. Otherwise $h^m = \widehat{h}_{it}^m$, and usable consumption and fertility solve

$$(1+\beta A)x^m + \chi n^m + u_t^m \widehat{h}_{it}^m = w_{it}, \tag{17}$$

$$\frac{\vartheta_t}{n^m} = \frac{\chi}{x^m} + \frac{\alpha\kappa}{\widehat{h}_{it}^m - \kappa n^m}. \tag{18}$$

The second equation prices the goods and housing taken from adults by another child. The appendix establishes a unique interior solution on this branch.

The value of relaxing the housing limit is the household's marginal value of space minus its user cost:

$$\zeta_{it}^{m} = \frac{\alpha x^{m}}{s^{m}} - u_{t}^{m} \geq 0. \tag{19}$$

It is zero without a binding limit. If only the owner's down-payment constraint binds, denote the wedge by $\zeta_{it}^{O,F}$. Combining the housing and fertility conditions gives

$$\frac{\vartheta_t x^{m}}{n^{m}} = \chi + \kappa(u_{t}^{m} + \zeta_{it}^{m}). \tag{20}$$

Thus a binding access constraint raises the effective cost of a child by $\kappa\zeta_{it}^{m}$. Appendix A.2 also gives the branch on which the old rental limit binds.

1.4 Positive steady states and transitions

A permanent change in desired fertility can move the economy between steady states. Prices and taxable gains are constant at either endpoint but can move during the transition.

Let $\bar{n}_t$, $\bar{h}_t^{Y}$, and $\bar{h}_t^{O}$ denote average fertility and housing per young and old household. Write stationary choices as functions of $(P, T; \vartheta)$ and set $\bar{h} = \bar{h}^{Y} + \bar{h}^{O}$. Market clearing and the government budget are

$$Y_t \bar{h}_t^{Y} + O_t \bar{h}_t^{O} = \bar{H}, \tag{21}$$

$$(Y_t + O_t)T_t = q\tau^{p} P_t \bar{H} + \ell_t O_t \bar{h}_t^{\text{sell}}, \tag{22}$$

where $\bar{h}_t^{\text{sell}}$ is housing sold by old owners per old household.

Proposition 1 (Steady-state accounting)**.** *Every positive steady state satisfies*

$$\bar{n}(P^*, T^*; \vartheta) = \frac{1}{\nu}, \qquad \ell^* = 0, \tag{23}$$

$$T^* = \frac{q\tau^{p} P^*}{2}\bar{h}(P^*, T^*; \vartheta), \qquad Y^* = O^* = \frac{\bar{H}}{\bar{h}(P^*, T^*; \vartheta)}. \tag{24}$$

Stationarity fixes average fertility at the replacement rate. Prices, transfers, tenure, individual fertility and housing, and the population scale can nevertheless differ across steady states.

Appendix A.3 gives conditions for local determination.

Suppose ϑ_t rises permanently at date zero. Let G_t be the old households' distribution over wealth, housing, and tenure, and write the state as $S_t = (Y_t, O_t, G_t, P_{t-1})$. Starting from the old steady state, a transition equilibrium is a sequence of prices, transfers, choices, and distributions that satisfies household optimality, the laws of motion, and the two market conditions at every date and converges to the new steady state. The two market conditions are

$$Y_t \bar{h}_t^{Y} + O_t \bar{h}_t^{O} = \bar{H}, \tag{25}$$

$$(Y_t + O_t)T_t = q\tau^{p} P_t \bar{H} + \ell_t O_t \bar{h}_t^{\text{sell}}. \tag{26}$$

The second equation rebates property-tax revenue and gains taxes paid by old sellers.

Figure 1 imposes new steady-state prices and transfers from date $J + 1$ onward. The induced state is recorded rather than imposed. Appendix A.4 establishes local determination of each finite system and verifies horizon stability; it does not establish an infinite-horizon existence or uniqueness result.

One permanent shock is sufficient for several cohorts to realize gains. The cohort law can be iterated as

$$Y_t = Y_0 \prod_{s=0}^{t-1} \nu \bar{n}_s, \qquad O_t = Y_{t-1}. \tag{27}$$

The first fertility response changes the next cohort, which later becomes old and produces another cohort. This propagation can move prices for several dates. Whenever $P_s > P_{s-1}$, old sellers realize $\ell_s = \tau^g(P_s - P_{s-1}) > 0$; no second exogenous shock is required.

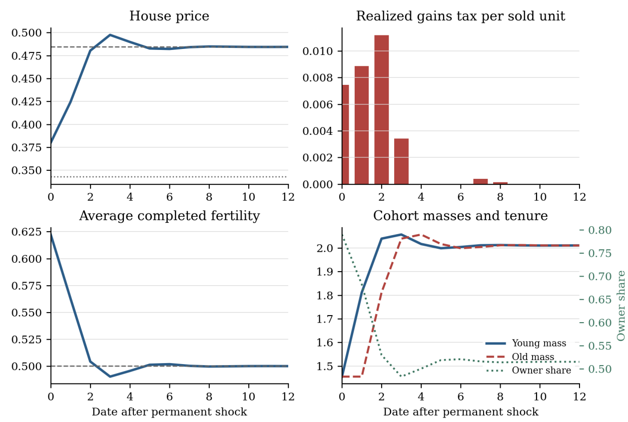

Figure 1: Transition after a permanent increase in the fertility preference. Lines mark steady-state prices and replacement fertility $1/\nu$.

1.5 Intergenerational reallocation and fertility

Consider transferring a marginal unit of housing from an old incumbent to a young owner, holding fertility, cohort sizes, tenure, prices, and other allocations fixed. The buyer's down-payment constraint binds and the other relevant choices are interior. Their marginal values are

$$MV^Y_{it} = q\,r_t + q\ell_{t+1} + \zeta^{O,F}_{it}, \qquad MV^O_{jt} = q\,r_t - \ell_t. \tag{28}$$

The common term qr_t values housing services. The incumbent avoids ℓ_t by retaining the unit, the buyer anticipates $q\ell_{t+1}$, and the financing constraint contributes $\zeta^{O,F}_{it}$.

A reallocation is a Pareto improvement for current households if, with transfers balanced across dates t and $t+1$, every household alive at t is weakly better off and at least one is strictly better off.

Proposition 2 (Pareto-improving reallocation)**.** *Under the conditions above and the two-date transfer ledger stated in Appendix A.5, the marginal-value difference is*

$$MV^Y_{it} - MV^O_{jt} = \zeta^{O,F}_{it} + \ell_t + q\ell_{t+1}. \tag{29}$$

Let $K_{ijt} \geq 0$ be the marginal resource cost of transferring title and occupancy. Purchase and tax payments are transfers. If $MV^Y_{it} - MV^O_{jt} > K_{ijt}$, a sufficiently small compensated reallocation is a Pareto improvement for current households.

At a steady state the condition is $\zeta^{O,F}_i > K_{ij}$; the gains-tax terms are transition-specific. The result does not rank economies with different populations.

A Derivations and numerical example

The log specification gives closed-form household policies on the branches used in the text. This appendix collects the derivations and describes the numerical transition.

A.1 Old-age problems

The old problems are (9)–(10). For owners, it is useful to define

$$A = 1 + \gamma + \omega_B, \qquad \varrho_t = q\,r_t - \ell_t, \qquad X_t = a + (P_t - \ell_t)H + T_t. \tag{30}$$

Here ϱ_t is the cost of retaining one unit and X_t values housing carried into old age at its after-tax sale price.

Lemma 1 (Old-age allocations)**.** *Suppose $r_t > 0$, $\varrho_t > 0$, $a + T_t > 0$, and $X_t > 0$. When housing is interior, renter and owner expenditure shares give*

$$(c^2_R, h^2_R, e_R) = \left(\frac{a+T_t}{A}, \frac{\gamma(a+T_t)}{Aq\,r_t}, \frac{\omega_B(a+T_t)}{Aq}\right), \tag{31}$$

$$(c^2_O, h^2_O, e_O) = \left(\frac{X_t}{A}, \frac{\gamma X_t}{A\varrho_t}, \frac{\omega_B X_t}{Aq}\right). \tag{32}$$

The owner row applies when

$$\frac{\gamma X_t}{A\varrho_t} < H, \qquad \omega_B \varrho_t > \gamma q P_{t+1}. \tag{33}$$

If the rental cap binds, set $Z^R = a + T_t - q\,r_t h^{\max}_R$. The allocation is then

$$h^2_R = h^{\max}_R, \qquad c^2_R = \frac{Z^R}{1+\omega_B}, \qquad e_R = \frac{\omega_B Z^R}{(1+\omega_B)q}. \tag{34}$$

Proof. For an interior budget $c^2 + qe + p_h h^2 = X$, the first-order conditions imply $p_h h^2 = \gamma c^2$, $qe = \omega_B c^2$, and $Ac^2 = X$. Setting $p_h = qr_t$ for renters and $p_h = \varrho_t$ for owners gives the two interior rows. At the rental cap, the residual Z^R is divided between consumption and the estate in the ratio $1 : \omega_B$. Substituting the owner row into $h^2_O < H$ and $e_O > P_{t+1}h^2_O$ gives (33). □

Fixing $h^2 = H$, $e = P_{t+1}h^2$, or both gives the remaining owner corners. Strict concavity makes each branch unique. On every branch used below, the wealth envelope is $V_{a,t} = 1/c^2_t$; on the interior owner branch, $V^O_{H,t} = (P_t - \ell_t)V^O_{a,t}$.

A.2 Young household solutions and tenure

Let λ^m_t be the current-budget multiplier, η^m_t the multiplier on the tenure-specific housing limit, and μ_t the owner down-payment multiplier. Using $x^m_t = c^m_t - \chi n^m_t$ and $s^m_t = h^m_t - \kappa n^m_t$, the consumption and fertility conditions are

$$\lambda^m_t = \frac{1}{x^m_t}, \tag{35}$$

$$\frac{\vartheta_t}{n^m_t} = \frac{\chi}{x^m_t} + \frac{\alpha\kappa}{s^m_t}. \tag{36}$$

When saving is interior, both tenure problems also satisfy $\lambda_t^m = \beta R_f V_{a,t+1}^m$.

The only additional branch needed for the numerical example is an old renter at the rental cap. The two possible saving loads and effective lifetime resources are

$$(k_{t+1}^m, \widetilde{w}_{it}^m) = \begin{cases} (\beta A, w_{it}), & \text{if old-age choices are interior,} \\ (\beta(1+\omega_B), w_{it} - q^2\, r_{t+1} h_R^{\max}), & \text{if } m = R \text{ and the old rental limit binds.} \end{cases} \tag{37}$$

The capped-renter row removes committed old-age rent from lifetime resources and removes old-age housing from the saving load. On either row, eliminating saving and old-age choices gives

$$\begin{aligned} \max_{x,s,n} \quad & (1+k_{t+1}^m)\log x + \alpha \log s + \vartheta_t \log n \\ \text{s.t.} \quad & (1+k_{t+1}^m)x + u_t^m s + (\chi + \kappa u_t^m)n = (1+k_{t+1}^m)x + \chi n + u_t^m h = \widetilde{w}_{it}^m. \end{aligned} \tag{38}$$

Thus renter and owner choices differ only through the user cost, effective resources, and housing limit.

On the same branches, the housing condition in consumption units is

$$\frac{\alpha x_t^m}{s_t^m} = u_t^m + \zeta_t^m, \qquad \zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \qquad \zeta_t^O = \frac{\mu_t}{\lambda_t^O}(1-\phi)P_t + \frac{\eta_t^O}{\lambda_t^O}. \tag{39}$$

This is (19) expressed in primitives. If the owner-size limit is slack, $\eta_t^O = 0$ and the owner's wedge is entirely due to the down payment.

Proposition 3 (Conditional household policies)**.** *Suppose young saving is interior. For owners, suppose next-period old retention and estate composition are interior. For renters, allow either an interior old housing choice or a strictly binding old rental limit. Write $B_{it}^m = 1 + k_{t+1}^m + \alpha + \vartheta_t$. If the current housing limit is slack, then*

$$x_t^m = \frac{\widetilde{w}_{it}^m}{B_{it}^m}, \qquad s_t^m = \frac{\alpha \widetilde{w}_{it}^m}{B_{it}^m u_t^m}, \qquad n_t^m = \frac{\vartheta_t \widetilde{w}_{it}^m}{B_{it}^m(\chi + \kappa u_t^m)}. \tag{40}$$

Gross choices are $c_t^m = x_t^m + \chi n_t^m$ and $h_t^m = s_t^m + \kappa n_t^m$. Saving on the interior old-renter, capped old-renter, and interior old-owner branches is

$$a_t'^{R,I} = \beta A x_t^R - qT_{t+1}, \tag{41}$$

$$a_t'^{R,C} = \beta(1+\omega_B)x_t^R - qT_{t+1} + q^2\, r_{t+1} h_R^{\max}, \tag{42}$$

$$a_t'^{O} = \beta A x_t^O - qT_{t+1} - [q(P_{t+1} - \ell_{t+1}) - \phi P_t] h_t^O. \tag{43}$$

The capped renter row applies when its implied old allocation satisfies $\gamma c_{t+1}^2 / h_R^{\max} > q\, r_{t+1}$; the interior row applies with the reverse inequality. If the current limit binds on a feasible branch, so that $\widehat{h}_{it}^m > 0$ and $\widetilde{w}_{it}^m > u_t^m \widehat{h}_{it}^m$, then $h_t^m = \widehat{h}_{it}^m$ and fertility is the unique root of

$$\frac{\vartheta_t}{n} - \frac{\alpha\kappa}{\widehat{h}_{it}^m - \kappa n} - \frac{(1+k_{t+1}^m)\chi}{\widetilde{w}_{it}^m - u_t^m \widehat{h}_{it}^m - \chi n} = 0 \tag{44}$$

on the feasible interval

$$0 < n < \min\left\{ \frac{\widehat{h}_{it}^m}{\kappa}, \frac{\widetilde{w}_{it}^m - u_t^m \widehat{h}_{it}^m}{\chi} \right\}. \tag{45}$$

Proof. The wealth envelope is $V_{a,t+1} = 1/c^2_{t+1}$, so the Euler equation gives $c^2_{t+1} = \beta R_f x_t$. For owners, old-age resources are $R_f a'_t - R_f \phi P_t h_t + (P_{t+1} - \ell_{t+1})h_t + T_{t+1}$. The old-age shares in Lemma 1 then imply that their date-t value is $\beta A x_t$. Combining this identity with the current budget cancels mortgage borrowing against repayment and gives the owner row of (38) and (43). The same calculation gives (41) for an interior old renter. At the old rental cap, resources net of committed rent are $Z^R_{t+1} = R_f a'_t + T_{t+1} - q\, r_{t+1} h^{\max}_R$. Equation (34) and the Euler equation give $qZ^R_{t+1} = \beta(1+\omega_B)x_t$, which yields the capped row of (37) and (42).

When housing is unconstrained, (39) and (36) give $s = \alpha x/u^m_t$ and $n = \vartheta_t x/(\chi + \kappa u^m_t)$. The lifetime budget then gives (40). At a binding limit, substitute $h = \widehat{h}$ and $x = (\widetilde{w}^m - u^m_t \widehat{h} - \chi n)/(1 + k^m_{t+1})$ into the fertility condition. The resulting left side is (44); it moves from $+\infty$ to $-\infty$ over (45) and has derivative

$$-\frac{\vartheta_t}{n^2} - \frac{\alpha\kappa^2}{(\widehat{h} - \kappa n)^2} - \frac{(1 + k^m_{t+1})\chi^2}{(\widetilde{w}^m - u^m_t \widehat{h} - \chi n)^2} < 0. \tag{46}$$

Hence the root exists and is unique. □

At a zero-saving or other old-age corner, the corresponding KKT conditions replace these formulas. Strict concavity still gives a unique policy within each tenure. The logistic taste then makes the owner share and aggregate choices continuous wherever both conditional values are finite.

A.3 Aggregation and positive steady states

Let F denote the distribution of observed young types $i = (y_i, b_i)$. The old-state distribution is over financial wealth a, housing H carried from young age, and tenure. For any bounded measurable test function ψ, its law of motion is

$$\begin{aligned}\int \psi \,\mathrm{d}G_{t+1} = \int \Big[&(1 - \pi^O_t(i))\psi(R_f a'^R_t(i), 0, R) \\ &+ \pi^O_t(i)\psi(R_f a'^O_t(i) - R_f \phi P_t h^O_t(i), h^O_t(i), O)\Big]\,\mathrm{d}F(i).\end{aligned} \tag{47}$$

Average fertility and young housing integrate the corresponding conditional policies with the same weights. Average old housing integrates the old policies over G_t. Average taxable sales are

$$\bar{h}^{\text{sell}}_t = \int \mathbf{1}\{m = O\}[H - h^2_t(a, H, O)]\,\mathrm{d}G_t. \tag{48}$$

Equations (21)–(22), the cohort laws, and (47) close the economy.

Proof of Proposition 1. The cohort laws do most of the work. Since today's old are yesterday's young, $O^* = Y^*$. A constant young cohort then requires $1 = \nu\bar{n}(P^*, T^*; \vartheta)$: average fertility must be at replacement. A constant real house price makes taxable gains zero. Housing clearing sets the common cohort mass at $Y^* = \bar{H}/\bar{h}$. Finally, the government budget is $2Y^*T^* = q\tau^p P^* \bar{H}$; substituting for Y^* gives the transfer equation in (24). Reversing these steps proves sufficiency once the stationary household distribution is constructed. □

For each house price, fiscal balance determines the transfer. A steady state is then a point on this fiscal locus at which fertility equals replacement.

Proposition 4 (Steady-state existence and local uniqueness)**.** *Let $\mathcal{D} = [\underline{P}, \overline{P}] \times [\underline{T}, \overline{T}] \subset \mathbb{R}^2_{++}$ be a rectangle on which the household policies are feasible, user costs are positive, and $\bar{n}$ and $\bar{h}$ are continuous. Define the fiscal map*

$$\Psi(P, T) = \frac{q\tau^p P}{2}\bar{h}(P, T; \vartheta). \tag{49}$$

Suppose $\Psi(P, \cdot)$ maps the transfer interval into itself and is a contraction in T, uniformly in P, with common modulus $L < 1$. Write $\mathcal{T}(P)$ for its fixed point. If

$$[\nu\bar{n}(\underline{P}, \mathcal{T}(\underline{P}); \vartheta) - 1][\nu\bar{n}(\overline{P}, \mathcal{T}(\overline{P}); \vartheta) - 1] < 0. \tag{50}$$

then a positive steady state exists in $\mathcal{D}$. If $\bar{n}$ and $\bar{h}$ are continuously differentiable near that state, it is locally unique when

$$J^* = \begin{bmatrix} \nu\bar{n}_P & \nu\bar{n}_T \\ -\dfrac{q\tau^p}{2}(\bar{h} + P\bar{h}_P) & 1 - \dfrac{q\tau^p P}{2}\bar{h}_T \end{bmatrix}_{(P^*, T^*; \vartheta)} \tag{51}$$

is nonsingular.

Proof. The contraction theorem gives one fiscally balanced transfer $\mathcal{T}(P)$ for each price. The common contraction modulus and continuity of Ψ make $\mathcal{T}(P)$ continuous. The sign change in (50) therefore yields a price at which fertility is $1/\nu$. Its fiscal fixed point completes the steady state. Nonsingularity of J^* gives local uniqueness by the implicit-function theorem. □

If J^* is nonsingular, the response to a permanent change in ϑ is

$$\begin{bmatrix} P^*_\vartheta \\ T^*_\vartheta \end{bmatrix} = -(J^*)^{-1} \begin{bmatrix} \nu\bar{n}_\vartheta \\ -\dfrac{q\tau^p P^*}{2}\bar{h}_\vartheta \end{bmatrix}. \tag{52}$$

The associated population-scale response is

$$\frac{\mathrm{d}\log Y^*}{\mathrm{d}\vartheta} = -\frac{\bar{h}_P P^*_\vartheta + \bar{h}_T T^*_\vartheta + \bar{h}_\vartheta}{\bar{h}}. \tag{53}$$

There is no general sign without restrictions on the demand and tenure elasticities.

A.4 Transition equilibrium

Suppose $\vartheta_t = \vartheta^0$ for $t < 0$ and $\vartheta_t = \vartheta^1$ for $t \geq 0$, and both values support regular positive steady states $\mathcal{E}^0$ and $\mathcal{E}^1$. The date-zero state is the old steady state: $S_0 = (Y^0, Y^0, G^0, P^0)$. In particular, $P_{-1} = P^0$ is the basis of date-zero old owners. No additional initialization shock is present.

The gains rule is equivalently the complementarity system

$$\ell_t \geq 0, \qquad \ell_t \geq \tau^g(P_t - P_{t-1}), \qquad \ell_t[\ell_t - \tau^g(P_t - P_{t-1})] = 0. \tag{54}$$

It is continuous but kinked. Within an appreciating regime its price derivative is τ^g; within a non-appreciating regime it is zero.

Fix a terminal date J and set prices and transfers from $J+1$ onward equal to their new steady-state values. A household that buys at J nevertheless keeps P_J as its tax basis, so its old-age gains tax is computed correctly. The state generated at $J+1$ is recorded rather than set equal to its steady-state value. Let z_J stack $(\log P_0, \ldots, \log P_J, \log T_0, \ldots, \log T_J)$ and order the residuals as the $J+1$ housing equations followed by the $J+1$ government equations. Recursively applying household policies, the distribution law, and cohort laws turns (26) into a finite system

$$\mathbf{R}_J(z_J; \varepsilon) = 0, \tag{55}$$

where ε scales the permanent change from ϑ^0 to ϑ^1 and the terminal steady state also varies with ε.

The gains rule has a kink at constant prices. To handle it, fix a candidate appreciation pattern $d = (d_0, \ldots, d_{J+1}) \in \{0,1\}^{J+2}$ and define the smooth branch

$$\ell_t^d = \tau^g d_t (P_t - P_{t-1}). \tag{56}$$

It agrees with the statutory rule when prices rise on dates with $d_t = 1$ and fall on dates with $d_t = 0$. Let $\mathbf{R}_J^d$ denote the residual system on this branch.

Proposition 5 (Local solution for a fixed terminal date)**.** *Suppose the old steady state is regular, the household active sets are strict, and the Jacobian $\partial \mathbf{R}_J^d / \partial z_J$ is nonsingular at the old steady state. If the directional response of $P_t - P_{t-1}$ is positive on dates with $d_t = 1$ and negative on dates with $d_t = 0$, then, for each fixed J, a sufficiently small positive shock has a locally unique terminally closed solution on branch d. This solution satisfies the statutory gains rule through date $J+1$.*

Proof. On a fixed branch, household policies and equilibrium residuals are smooth. At zero shock the constant old-steady-state path solves $\mathbf{R}_J^d = 0$. The implicit-function theorem therefore gives a locally unique solution for a small shock. The assumed price signs persist locally, so (56) coincides with the positive-part tax rule. □

The result is silent if a price response is tangent to the kink, and it does not rule out a solution on another branch. It is also a result for a fixed terminal date, not an existence theorem for the infinite-horizon transition. The horizon comparisons below show that the computed path is insensitive to extending J.

The cohort recursion in (27) lets one permanent shock move housing demand and capital-gains bases for several dates. The equilibrium path determines when appreciation, and hence taxable gains, occurs.

A.5 Compensated reallocation

For a constrained young owner, (39) gives the current-goods marginal value

$$MV_{it}^Y = \frac{\alpha x_{it}^O}{s_{it}^O} = q\, r_t + q \ell_{t+1} + \zeta_{it}^{O,F}. \tag{57}$$

For an old incumbent whose retention and estate-composition constraints are both slack, the housing first-order condition from (10) gives

$$MV_{jt}^O = \frac{\gamma c_{jt}^2}{h_{jt}^2} = q\, r_t - \ell_t. \tag{58}$$

Subtracting gives the decomposition in (29).

To translate the value gap into a welfare comparison, move one marginal unit, compensate the incumbent, finance the buyer at the market bond price, and return the incremental tax revenue. The only resource cost is the cost of changing title and occupancy.

Assumption 1 (Implementation)**.** The marginal purchase and any compensation can be financed at the bond price. The government rebates the incremental tax revenue and can make household-specific transfers at dates t and $t+1$. The buyer's old-age retention and estate constraints are slack. The added unit is sold at $t+1$ while the buyer's baseline old-age housing is held fixed. Fertility, tenure, cohort masses, and all other allocations are unchanged. The transfer cost K_{ijt} is measured in date-t goods.

Proof of Proposition 2. Transfer $\epsilon > 0$ units from the old incumbent to the young buyer. The baseline has the incumbent retain the unit, transfer it at death with basis P_{t+1}, and have the estate sell it at date $t+1$ without gains tax. The following ledger values each change in date-t consumption goods and divides by ϵ:

Agent	Change relative to the baseline	Net value
Incumbent j	Receives P_t, pays ℓ_t, and gives up a unit whose net service-and-estate value, after property tax, is $P_t - \ell_t$ at the interior retention choice.	0
Buyer i	Pays P_t using market-rate bridge finance, pays $q\tau^p P_t$, receives current services worth MV_{it}^Y, and the added unit is sold next period for $q(P_{t+1} - \ell_{t+1})$.	$\zeta_{it}^{O,F}$
Government	Property-tax revenue is unchanged; the incumbent sale and the buyer's later sale yield revenue absent from the baseline estate sale.	$\ell_t + q\ell_{t+1}$
Lenders	Bridge and mortgage funds are supplied at the bond price.	0
Real resources	Transferring title and occupancy uses K_{ijt} units of current goods.	$-K_{ijt}$
Total		$\zeta_{it}^{O,F} + \ell_t + q\ell_{t+1} - K_{ijt}$

For the buyer row, substituting $MV_{it}^Y = q\, r_t + q\ell_{t+1} + \zeta_{it}^{O,F}$ and $q\, r_t = P_t + q\tau^p P_t - qP_{t+1}$ gives the displayed net value. For the incumbent row, the retention first-order condition makes the marginal unit's net service-and-estate value equal to its after-tax sale proceeds: $MV_{jt}^O - q\tau^p P_t + qP_{t+1} = P_t - \ell_t$. These identities count the current housing service and the transfer of title once each.

Assumption 1 assigns the added unit to the date-$(t+1)$ sale while holding the buyer's baseline old-age housing fixed. Slack old-age constraints make this perturbation feasible, and the old-age first-order conditions make it welfare-neutral to first order.

If total net value is positive, date-t and date-$(t+1)$ transfers can leave both affected households weakly above baseline and one strictly above it. Smoothness and the strict first-order inequality imply the same conclusion for a sufficiently small finite transfer. At a steady state the two gains terms are zero, so the condition becomes $\zeta_i^{O,F} > K_{ij}$. □

The Pareto statement depends on the financing and transfers in Assumption 1. Without them, the value gap remains a useful decomposition but need not be implementable. The proposition also fixes population. Comparing paths with different populations requires a social criterion for economies with different people; see Golosov et al. (2007) and Pérez-Nievas et al. (2019).

The tax treatment of rental intermediaries affects incidence but not the decomposition. If an intermediary also pays ℓ_{t+1} upon resale, its zero-profit rent is $r^L_t = r_t + \ell_{t+1}$. The young owner's private current-goods cost becomes $q\, r^L_t$, while the incumbent's retention cost can be written $q\, r^L_t - \ell_t - q\ell_{t+1}$. Their difference remains $\zeta^{O,F}_{it} + \ell_t + q\ell_{t+1}$.

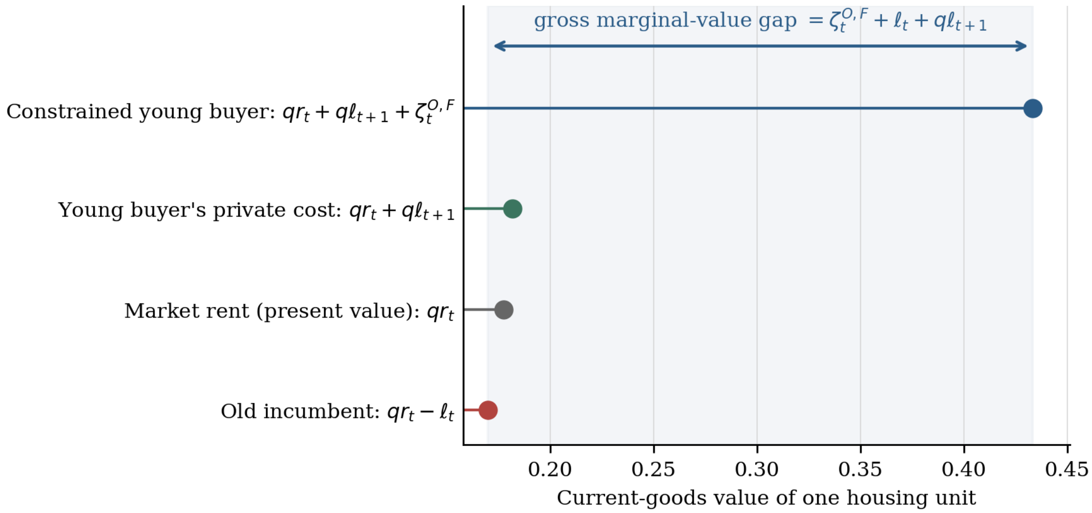

Figure 2: Shadow-value decomposition on the computed path. The financing wedge raises the young buyer's value; realization taxes lower the old incumbent's value and raise the buyer's anticipated private cost. The displayed object is the gross marginal-value gap. Under the ledger in Proposition 2, the reallocation is a Pareto improvement for current households when this gap exceeds the real transfer cost.

A.6 Housing access and fertility

Relaxing a binding housing limit has two effects. It gives the family more space, which makes children less costly, but it also requires the household to pay for more housing, which leaves fewer goods for children. The derivative below compares these two forces.

Suppose $h = \widehat{h}$ binds and fertility remains interior. Define

$$x = \frac{\widetilde{w}^m_{it} - u^m_t \widehat{h} - \chi n}{1 + k^m_{t+1}}, \qquad s = \widehat{h} - \kappa n, \tag{59}$$

and collect the positive terms in

$$\Delta = \frac{\vartheta_t}{n^2} + \frac{\chi^2}{(1 + k^m_{t+1})x^2} + \frac{\alpha\kappa^2}{s^2} > 0. \tag{60}$$

Implicit differentiation of the binding-fertility condition gives

$$\frac{\mathrm{d}n}{\mathrm{d}\widehat{h}} = \frac{\alpha\kappa/s^2 - \chi u^m_t/[(1 + k^m_{t+1})x^2]}{\Delta}. \tag{61}$$

On the interior old-age branch, the derivative is nonnegative exactly when

$$\chi \leq \frac{(1 + \beta A)\kappa(u^m_t + \zeta^m_{it})^2}{\alpha u^m_t}. \tag{62}$$

Derivation of (61) *and* (62). Let the left side of (44) be $F(n, \widehat{h})$. Direct differentiation gives

$$
\begin{aligned}
F_n &= -\Delta < 0, && (63)\\
F_{\widehat{h}} &= \frac{\alpha\kappa}{s^2} - \frac{\chi u_t^m}{(1+k_{t+1}^m)x^2}. && (64)
\end{aligned}
$$

The implicit-function theorem gives $\mathrm{d}n/\mathrm{d}\widehat{h} = -F_{\widehat{h}}/F_n$, which is (61). Its numerator is the difference between the space benefit and the goods-resource cost described above. At a constrained allocation, (39) implies

$$
\frac{x}{s} = \frac{u_t^m + \zeta_{it}^m}{\alpha}. \tag{65}
$$

Substituting this ratio into $F_{\widehat{h}} \geq 0$ gives the condition in (62) on the interior old-age branch, where $k_{t+1}^m = \beta A$. On the capped old-renter branch the same derivation uses $k_{t+1}^R = \beta(1+\omega_B)$. □

When the down-payment constraint alone binds,

$$
\begin{aligned}
\widehat{h}_{it}^O &= \frac{b_i}{(1-\phi)P_t},\\
\frac{\partial n}{\partial \phi} &= \frac{\partial n}{\partial \widehat{h}} \frac{b_i}{(1-\phi)^2 P_t},\\
\frac{\partial n}{\partial b_i} &= \frac{\chi}{(1+k_{t+1}^O)x^2\Delta} + \frac{\partial n}{\partial \widehat{h}} \frac{1}{(1-\phi)P_t}.
\end{aligned} \tag{66}
$$

The first expression isolates the access channel. Liquid wealth also has a direct resource effect, shown by the first term in its derivative. A funded policy additionally changes prices, transfers, and tenure, so its aggregate effect must be computed in equilibrium.

A.7 Mixed-tenure example

The numerical example gives observed types common income and liquid wealth and retains the logistic ownership preference in (11). It computes the two steady states and a terminally closed transition between them. Table 1 lists the parameters.

Table 1: Parameters of the mixed-tenure construction

Object	Parameter	Value	Object	Parameter	Value
Bond price	q	0.50	Mortgage share	ϕ	0.80
Property tax	τ^p	0.05	Gains-tax rate	τ^g	0.20
Space weight	α	0.40	Space per child	κ	0.50
Goods per child	χ	0.15	Young discount	β	0.40
Old housing weight	γ	0.30	Estate weight	ω_B	0.40
Income	y	1.00	Liquid wealth	b	0.06
Household formation	ν	2.00	Housing stock	$\bar{H}$	2.00
Rental limit	$h_R^{\max}$	0.45	Owner limit	$h_O^{\max}$	2.00
Owner intercept	$\bar{\xi}$	−0.15	Taste scale	σ_ξ	0.12

At date zero, ϑ rises permanently from 0.35 to 0.50. Solving (24) gives Table 2.

Table 2: Positive steady-state endpoints

Object	Initial steady state	Final steady state
Fertility preference ϑ	0.3500	0.5000
Average fertility $\bar{n}$	0.5000	0.5000
House price P	0.3427	0.4844
Per-household transfer T	0.00589	0.00602
Young owner share	0.7486	0.5149
Renter fertility	0.3527	0.4368
Owner fertility	0.5495	0.5596
Young renter housing	0.4500	0.4500
Young owner housing	0.8754	0.6193
Old renter housing	0.4500	0.4500
Old owner housing	0.6582	0.4649
Young and old cohort mass	1.4553	2.0103

Average fertility is $1/\nu = 0.5$ at both endpoints, but conditional fertility, tenure, housing, prices, and population scale differ. Both roots are locally regular: the Jacobian determinants in $(\log P, \log T)$ are -0.00466 and -0.00433.

On impact, average fertility rises to 0.6223 and the young cohort expands one period later. The price rises from 0.3427 to 0.3800, reaches 0.4975 at date 3, and then approaches 0.4844. Gains taxes are positive on dates 0–3 and 7–8. Thus one preference change generates taxable gains for several cohorts even though gains are zero in both steady states.

Both tenures have positive mass. The young renter and owner housing limits bind, and the owner's limit is financial. The old rental limit binds, while old owners are interior. The maximum scaled equilibrium residual is 6.51×10^{-10}; extending the horizon from 24 to 28 changes the first nine dates by at most 1.78×10^{-15}.

As a numerical check, evaluate the fiscal map on a 41×31 grid over $P \in [0.25, 0.60]$ and $T \in [0.005, 0.02]$. Every sampled value lies in the transfer interval, and the largest sampled absolute derivative with respect to T is 0.00588 at the initial parameter and 0.00560 at the final one. The replacement residual has opposite signs at the two price boundaries in both cases. These calculations support, but do not prove, the uniform conditions in Proposition 4.

References

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